Yang Lu

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Summary

Software engineer with about 5 years of experience building and shipping AI agent systems into production. Hands-on with LLMs, MCP, function calling, RAG, and multi-model evaluation across the full stack. Open-source contributor to Hermes Agent (NousResearch) — reliability, evaluation, and tooling improvements.

Experience

Independent AI Systems Engineer
02/2023 – Present

AI Integration Solutions | Remote

- Designed and shipped production AI agent systems — multi-step workflows wiring together LLMs, MCP servers, internal APIs, and data sources with human-in-the-loop controls.
- Built agent reliability infrastructure: dead-letter queues, 429 exponential backoff with reconnect replay, message routing — 10 open-source PRs to NousResearch/hermes-agent.
- Created LLM evaluation framework with LLM-as-judge relevance scoring, async parallel evaluation across multiple providers, and latency benchmarking.
- Deployed a LangChain RAG application with document ingestion, Hugging Face embeddings, Chroma retrieval, and OpenAI-compatible chat generation — live on Hugging Face Spaces.

Full-Stack Developer
07/2023 – 07/2024

EduTrigger | Toronto, ON

- Developed full-stack features in NextJS, React, and TypeScript, consuming REST APIs and working with PostgreSQL schema on an Open edX platform.
- Deployed and managed Docker-based infrastructure; implemented CSS, image, and configuration updates.

Back-End Developer
07/2016 – 07/2020

Alaia Technologies | Toronto, ON

- Extended and maintained Java/Spring Boot batch loaders consuming external REST APIs — 150K records/day into MySQL + Elasticsearch via Quartz scheduler; rewrote SQL queries for performance.
- Performed code reviews for junior engineers, used Jira daily for ticket-based Agile delivery, and maintained documentation in Confluence.

Projects

AI Agent Infrastructure & Reliability
Python, Hermes Agent Framework (NousResearch)

- 10 open-source PRs: gateway dead-letter queues, 429 exponential backoff with reconnect replay, CJK search tokenization.

LLM Eval Toolkit
Python, httpx/asyncio, OpenRouter API, Docker, GitHub Actions CI/CD

- Open-source multi-model LLM evaluation — async parallel LLM-as-judge scoring, relevance assessment, and latency benchmarking.

LangChain RAG Application
Python, LangChain, Chroma, Hugging Face Embeddings

- End-to-end RAG pipeline: ingestion → embeddings → Chroma retrieval → OpenAI-compatible chat generation. Deployed on Hugging Face Spaces.

Skills

AI/ML: LangChain, RAG, LLMs, MCP, function calling, agent frameworks, LLM-as-judge evaluation, prompt engineering, embeddings, vector DBs (Chroma), OpenRouter API
Languages: Python, Java, TypeScript, SQL
Backend: FastAPI, Spring Boot, Hibernate (JPA/ORM), REST APIs, GraphQL, PostgreSQL, MySQL
DevOps: Docker, GitHub Actions CI/CD, Linux/macOS server administration
Tools: Git, Pytest, Jira, Confluence, Agile/Scrum

Education

Queen's University
2016

Bachelor of Computing — Teradata Scholarship

- Certifications: HuggingFace MCP (10/2025), DeepLearning.ai Data Engineering (06/2025)